Lab 1: GNSS Plot Upload + Related Analysis Questions

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Assignment Overview

This assignment covers GPS data collection and analysis using the BU-353N sensor, including position accuracy evaluation, error characterization, and comparison between open area and occluded environments for both stationary and moving data.

Question 1

Please state three major environmental/systemic sources that might increase error in GPS. Which may have an effect on your wide open data sets? Which may have an effect on your occluded data sets?

Answer:

- Water in the form of rain/droplets/fog
- Trees might affect GPS signals
- Being stuck between buildings may reflect/bounce GPS signals
- Walls can prevent GPS signals from being received

Question 2

Please upload your scatterplot as a JPG or PNG (open area and occluded) of northing vs. easting, with the centroid subtracted from each dataset. As a reminder, a good plot should include:

- Axis labels
- Appropriate units
- Legible text

Answer:

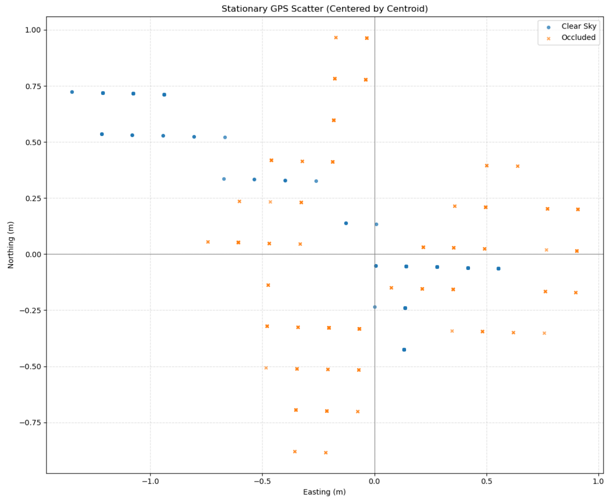


Figure 1: Scatterplot of northing vs. easting for stationary data (open area and occluded) with centroid subtracted

Question 3

Please upload your scatterplots as JPG or PNG (open area and occluded) of altitude vs. time. As a reminder, a good plot should include:

- Axis labels
- Appropriate units
- Legible text

Answer:

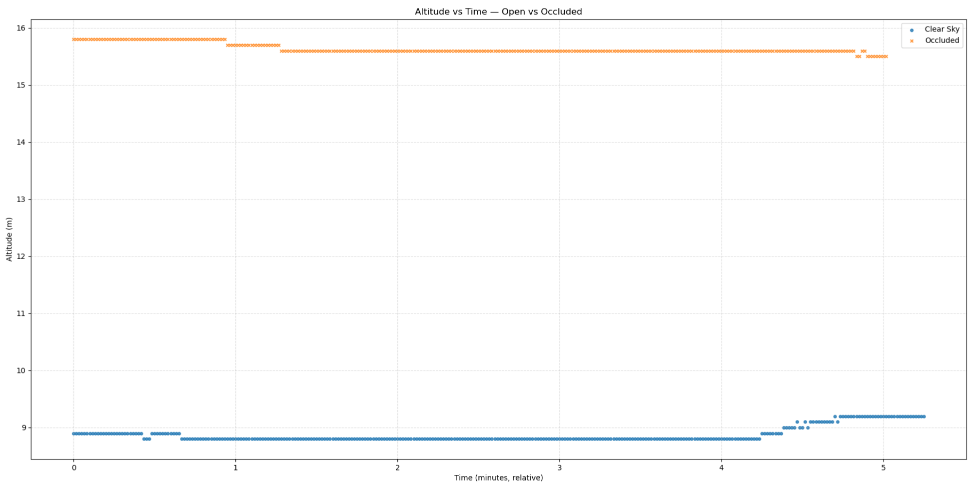
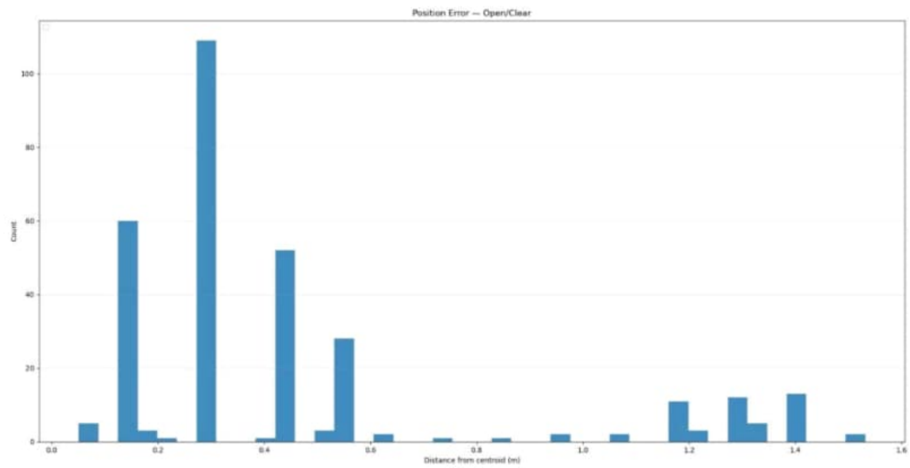


Figure 2: Altitude vs. time plots for stationary data (open area and occluded)

Question 4

Please upload your histogram plots (open area and occluded stationary data) of error from your "known" stationary position.

Answer:



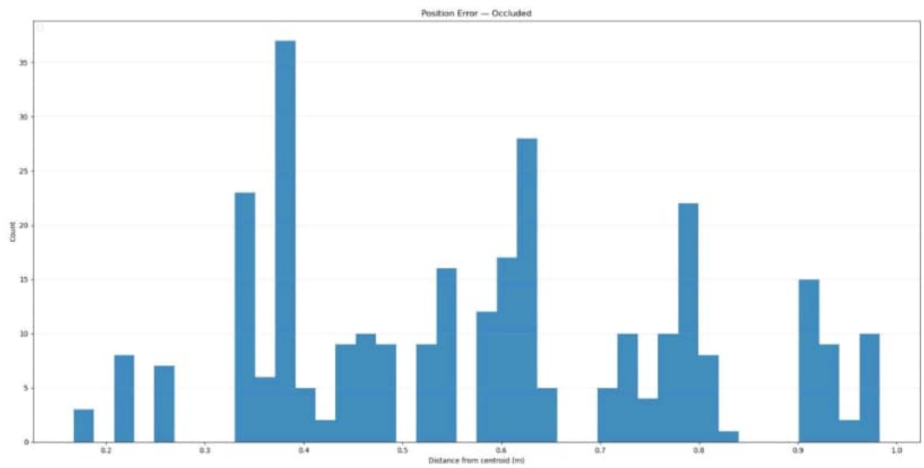


Figure 3: Histogram of position error for stationary data in open area and occluded environments

Question 5

Using your northing vs. easting stationary data and your "known" location, what is your error/deviation? How did you calculate this?

Please remember a completely correct answer will include a short calculation and appropriate units.

Answer:

I measured the euclidean distance from the centroid, not from a known reference location in order to measure the precision of the GPS device. This would measure how consistent the readings are as opposed to the other method of using a known reference location to find how close the GPS position is to the actual position.

Question 6

Do your values for error/deviation in northing vs. easting data and your altitude data make sense 1) given your hdop values, and 2) given general precision for civilian GPS? Why or why not? How did the measurements change from a wide open area to an occluded one?

Answer:

The HDOP (Horizontal dilution of precision) values were around 0.75-2, which makes the expected horizontal error values to be 1.5-3 meters, which our data mostly meets. Our VDOP values are usually twice our HDOP values, which would be enough to give us errors of 3-5 meters. In open vs occluded areas, our horizontal data values are much more accurate than vertical data, which has bigger errors/deviations. This means that moving from an open to occluded area worsens accuracy.

Question 7

Please upload your scatterplot of northing vs. easting for walking data

Answer:

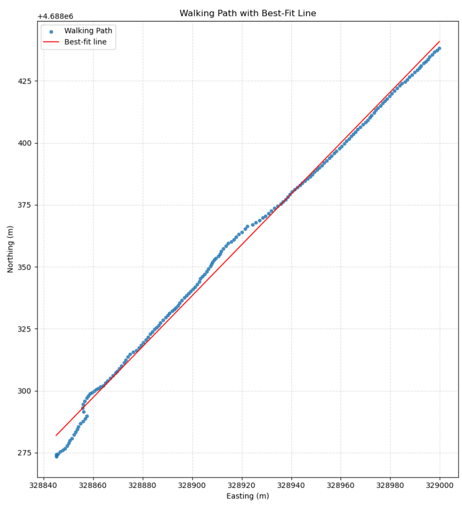


Figure 4: Scatterplot of northing vs. easting for walking data with line of best fit

Question 8

Please upload your scatterplot of altitude vs. time for walking data.

Answer:

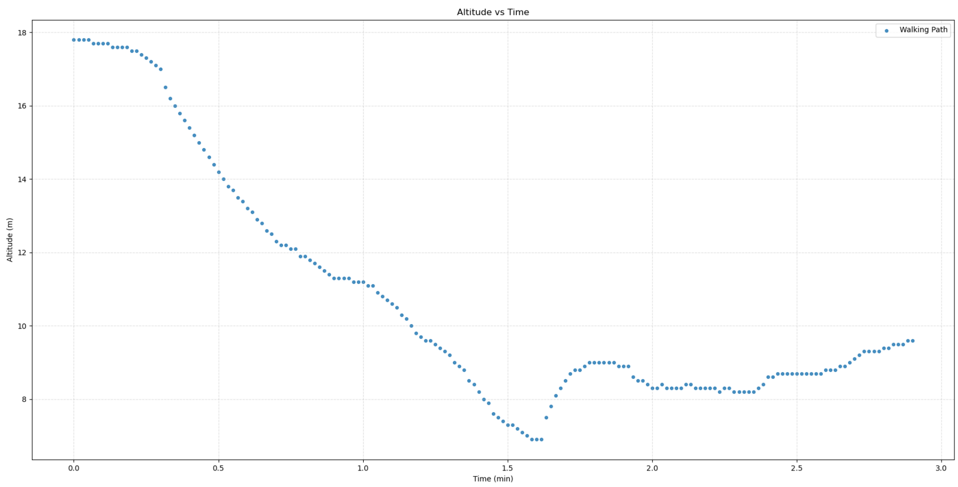


Figure 5: Altitude vs. time plot for walking data

Question 9

Assume you walk in a perfectly straight line (not a great assumption, but we'll roll with it!) What is the error/deviation (give a number) from a line of best fit to your walking northing vs. easting data? How did you calculate this error? Please remember a completely correct answer will include a short calculation and appropriate units.

Answer:

The way we can use to find the error/deviation in GPS data vs straight line (assumed) data is that we can measure the perpendicular distance from our line of best fit and our actual GPS points using tools like MATLAB/matplotlib which would give us deviations. After this, we get the deviations and we can take the average of all these deviations (which are perpendicular distances), which would give us an average error.

My analysis script prints out the average error using this exact working principle.

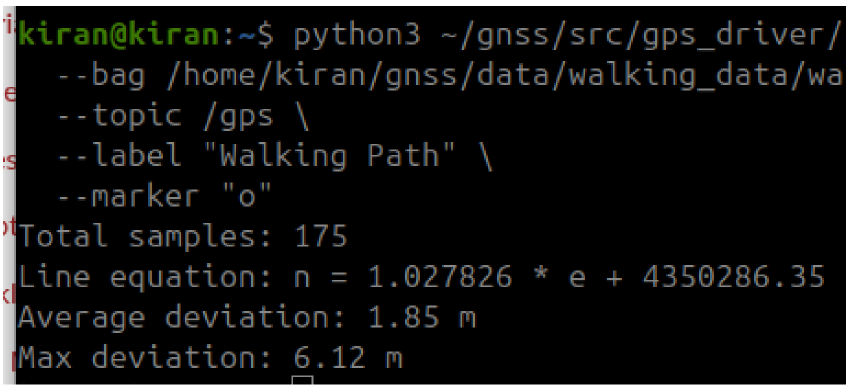


Figure 6: Screenshot showing average error calculation output

I'll show on pen and paper what that would mathematically mean.

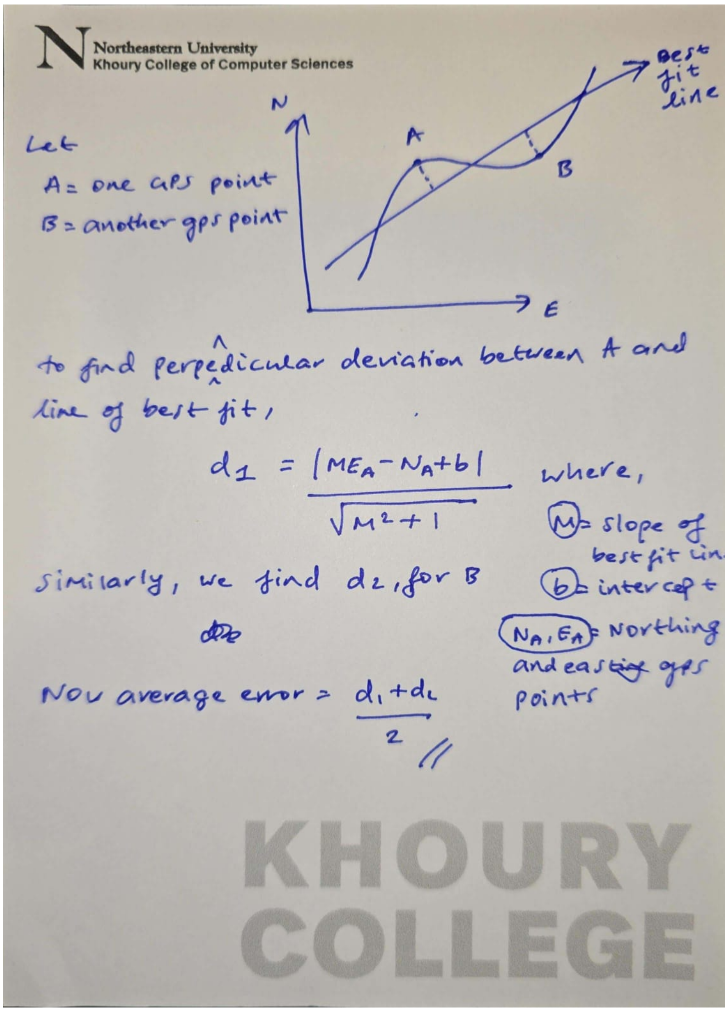


Figure 7: Mathematical working for error calculation

Question 10

How do your values for error in northing vs. easting walking data and your altitude data compare to 1) your hdop values, 2) general precision for civilian GPS, and 3) your stationary data sets? Can you explain any discrepancies?

Answer:

Comparing the error values for moving northing vs easting and altitude data, comparing it to HDOP values:

- Our dots on the best of fit line are around 1.5-3 meters off from expected, and our HDOP was in the range of 0.75 to 2, which is in line with our data.

Comparing values with general civilian GPS:

- Civilian grade GPS give us around 2-4 meters of accuracy, with our data here showing an accuracy of 1.5 meters, which indicates excellent satellite position, a clear sky, and clear walking path, all of which are best case scenarios.

Comparing with stationary data sets:

- The scatterplot range is within 1-1.5 meters, which means it is more accurate than moving data, which could be explained by transmission and receiving delays when walking from one place to another, or maybe signal processing delays that may in the order of milliseconds.